CO3 / RECONSTRUCTION

CO4 / GENERATION

GENERATIVE AI · ASSIGNMENT 02

From reconstruction to generation

A practical comparison of four image models

ONE STUDY · TWO TASKS · FOUR MODELS

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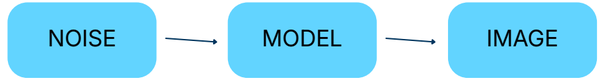
MIT Academy of Engineering

PART A / CO3



PART B / CO4

Generate



DCGAN + Stable Diffusion

STUDY FOCUS Sharpness · Detail · Diversity · Control

Four models. Two different jobs.

The first pair rebuilds an input; the second pair creates new images.

01 CO3 · RECONSTRUCT

Autoencoder

Compresses an image into a 16-value code, then rebuilds it.

IMAGE › ENCODER › z = 16 › DECODER

STRENGTH

Simple, fast; useful for compression and denoising.

TRADE-OFF

Latent space is unstructured; random codes give blobs.

USE

Compression · denoising · anomaly detection

Best reconstruction MSE in our run: 0.0108.

02 CO3 · RECONSTRUCT

Variational AE

Learns a Gaussian latent distribution; samples z and decodes it.

IMAGE › μ / σ › SAMPLE z › DECODER

STRENGTH

Smooth latent space supports sampling and interpolation.

TRADE-OFF

KL regularisation trades detail for structure; blurrier outputs.

USE

Sampling · interpolation · representation learning

MSE 0.0145; random z gives recognisable garments.

03 CO4 · GENERATE

DCGAN

Generator makes images; discriminator judges real vs fake.

NOISE z › GENERATOR › FAKE IMG › JUDGE

STRENGTH

Sharp samples in one generator pass.

TRADE-OFF

Training can be unstable; mode collapse; no prompt control.

USE

Image synthesis · data augmentation · style generation

Varied 28×28 samples; some malformed.

04 CO4 · GENERATE

Stable Diffusion

Denoises random latent noise step by step, guided by text.

PROMPT › TEXT ENC. › U-NET › VAE DEC.

STRENGTH

High realism, fine detail and text guidance.

TRADE-OFF

Many denoising steps; compute-heavy; can create artifacts.

USE

Text-to-image · inpainting · design

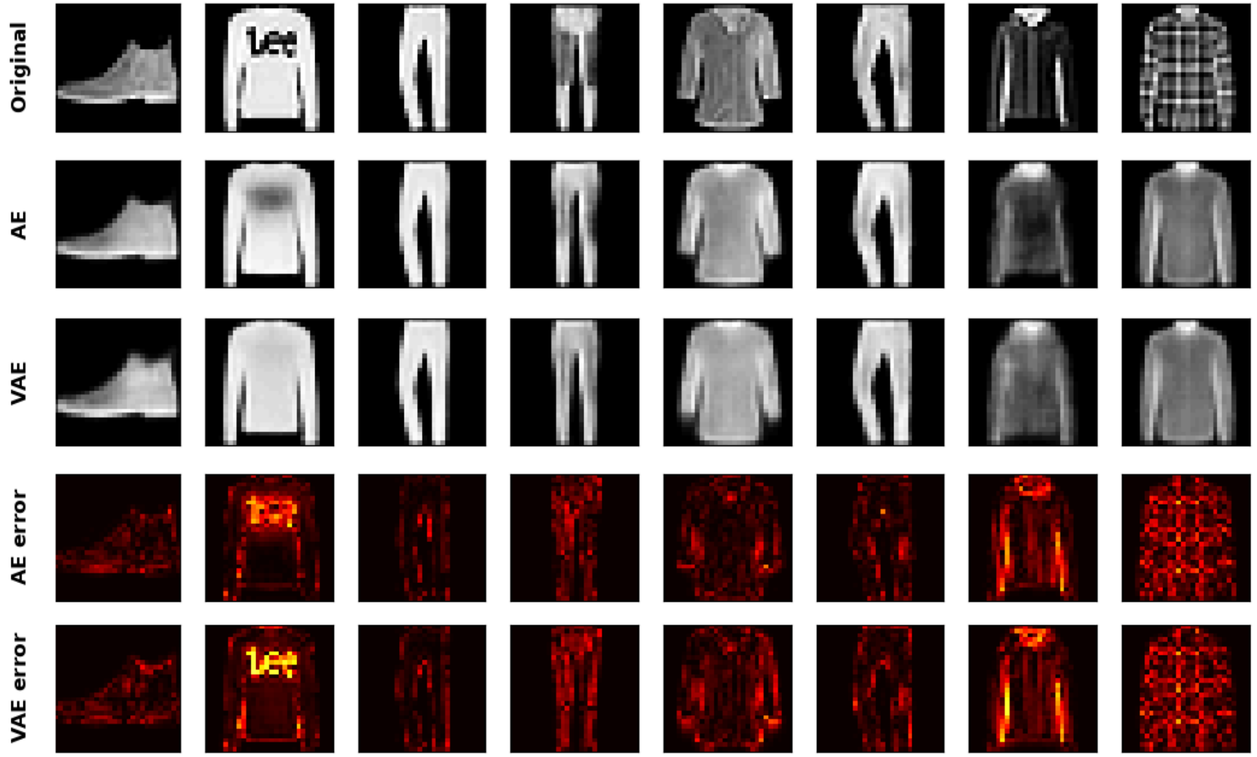
512×512 colour images with strong prompt match.

Architecture cue: AE = encoder + bottleneck + decoder · VAE = mean / log-variance + sampling · DCGAN = generator + discriminator · Diffusion = text encoder + U-Net + VAE decoder

Reconstruction is an accuracy–sampling trade-off.

Same eight Fashion-MNIST test images · 16-value bottleneck · Colab T4

INPUT → RECONSTRUCTION → ERROR MAP



The outlines survive; logos, plaid, zips and fine texture fade.

0.0108

AE / MEAN MSE

0.0145

VAE / MEAN MSE

34% HIGHER MSE FOR VAE

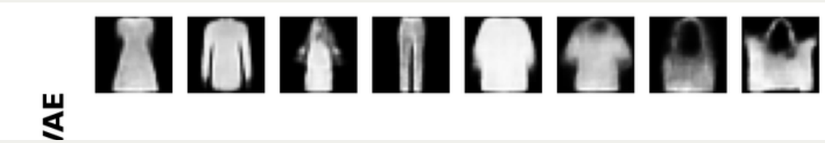
What the outputs show

- Both preserve garment silhouettes, but smooth fine detail.
- VAE has higher error on all 8 images; it removes the “Lee” logo.
- The plaid shirt becomes almost flat grey in both models.

RANDOM LATENT VECTORS

AE → blobs

VAE → garments



Fairness note: AE = 10 epochs / MSE; VAE = 20 epochs / BCE + KL. Only 8 images were scored, so the MSE gap is indicative, not a statistical benchmark.

Generation: diversity versus prompt control.

DCGAN learns from Fashion-MNIST; Stable Diffusion starts from a pretrained text-to-image model.

MODEL 01 DCGAN

64 generated samples



20 EPOCHS

326.7 s
training time

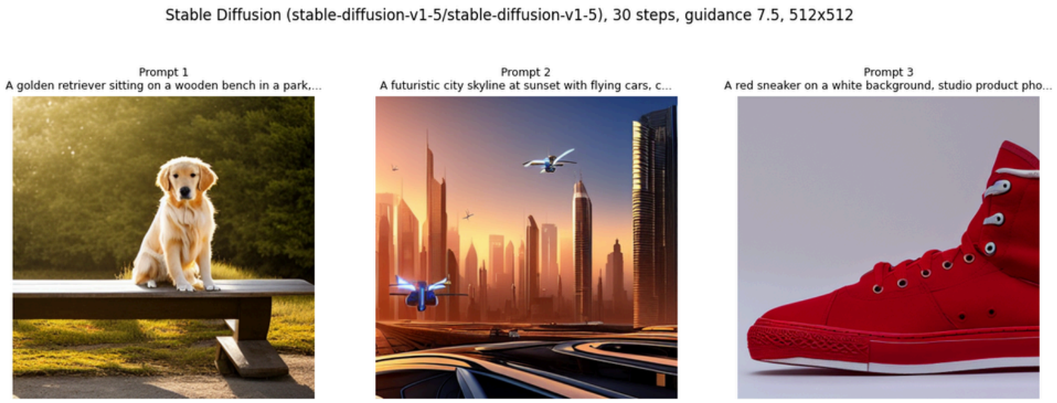
2.33 M
generator parameters

28 × 28 · grayscale

Varied clothing; no obvious mode collapse in this grid. Some samples are malformed, and the model cannot follow a prompt.

MODEL 02 Stable Diffusion v1.5

3 fixed prompts



512×512
COLOUR OUTPUT

30 steps
GUIDANCE 7.5

6.1–9.3 s
PER IMAGE / T4

Realistic fur, lighting and city detail; follows the prompt. “Flying cars” resemble drones, and the sneaker is cropped.

DCGAN = one-pass, compact, uncontrolled | DIFFUSION = prompt-guided, detailed, compute-intensive

Not a controlled benchmark: the two models were evaluated on different datasets and output resolutions.

What the experiments taught us

The result is not one “best” model — it is a set of trade-offs.

01

Reconstruction ≠ generation

AE has lower MSE. VAE gives a smoother latent space that can generate recognisable samples.

02

Inspect the images

GAN loss curves alone do not prove quality. Look for sample variety and mode collapse.

03

Detail has a cost

Diffusion offers stronger realism and prompt control, but needs more compute and steps.

AI USAGE / VERIFICATION

Claude supported planning, research, notebook drafting, debugging and report writing.

PROMPT
Plan + draft
→
RUN
Execute code
→
CHECK
Inspect output
→
FIX
Correct claims

Checks caught two mismatches: Fashion-MNIST (not MNIST) and 10 vs 20 AE/VAE epochs. A VAE helper-name collision was also fixed.

TEAM CONTRIBUTIONS

KRISHNA — Documentation lead
Report, PPT, output analysis, figures and final assembly.

MANMATH — Research & code lead
Model research, comparison, notebooks, debugging and prompt log.

VISHAL — Execution & evidence lead
Colab runs, screenshots, observations and runtime recording.

THEORY CHECKED AGAINST

Kingma & Welling (2014) · Goodfellow et al. (2014) · Radford et al. (2016) · Ho et al. (2020) · Rombach et al. (2022) · Hinton & Salakhutdinov (2006)

ALL THREE MEMBERS REVIEWED THE FOUR MODELS AND BOTH ACTIVITIES